Assignment - 10

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Audio Mixing and Processing using Audacity

Aim

To perform audio processing using Audacity by importing songs, separating vocals and music, segmenting the audio, and mixing different tracks to create a final composite audio output.

Software Used

- Audacity (Audio editing software)
- OpenVINO AI Effects (for voice and music separation)

Source Media

Two audio tracks were used:

- **Barandaye Roddur**
- **Apna Time Aayega**

These were downloaded from online sources and stored locally in MP3 format.

Procedure

1. Importing Audio Files

- Opened Audacity
- Used **File** → **Import** → **Audio**
- Loaded both songs into the workspace as separate tracks

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2. Trimming the Audio

- Selected a **30-second segment** from each song.
- **Trimmed** the unwanted portions.
- Ensured both songs had equal duration segments for uniform processing

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3. Separating Voice and Music

- Applied **OpenVINO AI Effects** on each trimmed track
- Generated:
    - **Vocal track (voice)**
    - **Instrumental track (music)**
- Result: Each song was split into two components:
    - Song 1 → Voice + Music
    - Song 2 → Voice + Music

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4. Arranging and Mixing Tracks

- Arranged the clips as a mix:

  Voice of song1 + music of song2 followed by voice of song2 + music of song1.

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5. Enhancing Audio Quality

- Adjusted volume levels.
- Ensured balanced output between both songs

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6. Exporting Final Output

- Used **File → Export → Export as MP3**
- Saved the final mixed audio file for submission

Result

A final mixed audio track was successfully created by combining vocals and music segments from two different songs in a structured sequence.

Conclusion

This experiment demonstrated the practical use of Audacity for audio editing and mixing. The process involved importing audio, trimming, separating vocals and music using AI-based tools, splitting tracks, and arranging them creatively. It provided hands-on experience in digital audio processing and enhanced understanding of track manipulation and synchronization.